

Tristan Varner

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Experience

IBM

Forward Deployed Engineering Intern

Durham, NC

May 2026 – Aug 2026

- Built the Python ingestion and retrieval layer for an enterprise RAG agent used by on-call Forward Deployed Engineers, including concurrent crawling, multi-format parsing, and metadata pipelines, contributing to a **97% pass rate** across an **800+ question** evaluation set.
- Indexed **200,000+ IBM Z documents** with 8 concurrent workers operating at the API's 32 requests/second limit, in a pipeline designed to scale to IBM's full **2.5M+ document corpus**.
- Designed a custom **MCP context server** exposing IBM product documentation to watsonx Orchestrate and internal Forward Deployed Engineering tools, giving agents grounded product context for customer troubleshooting.
- Partnered with senior FDEs on enterprise watsonx deployments, extending the agent with log analysis and configuration validation to diagnose recurring customer issues across IBM Z environments.

Firebirds Wood Fired Grill

Restaurant Manager

Durham, NC

May 2023 – May 2026

- Ran front- and back-of-house operations for a restaurant serving **600–800 covers** and **\$20K+ in daily sales**, leading 30+ employees across labor scheduling, inventory, cash management, and guest experience.
- Held ticket times under **20 minutes** at peak volume while managing food cost and labor to budget targets.

Projects

Cyclops – Smart Glasses Vision Platform

Python, FastAPI, Swift, PyTorch, CUDA

- Built an end-to-end vision platform streaming first-person video from Ray-Ban Meta glasses through an iPhone to a GPU workstation at **~12 FPS** for real-time computer vision.
- Developed World Builder, a spatial-computing pipeline that estimates depth and camera motion from live wearable footage and persists spatial data for reconstructing and revisiting real environments.
- Designed a modular “cartridge” architecture that swaps independent capabilities (spatial reconstruction, object memory, document understanding) on the same platform without rebuilding the core system.

Gary – Autonomous Software Engineering Platform

Python, FastAPI, MCP, Ollama, SQLite, TypeScript

- Built a local-first autonomous software engineering platform where Planner, Coder, and Reviewer agents run structured **implement–test–review–fix** loops against real codebases.
- Designed a backend-authoritative execution architecture, with role-based permissions, sandboxed filesystem access, and persistent state held outside model control, so agents can modify real projects safely.
- Tested across multiple Llama and Qwen models (7B–14B) to reduce single-model bias; supports automatic failure recovery, persistent sessions, and a packaged VS Code extension, backed by **1,100+ passing backend tests**.

Education

The University of North Carolina at Charlotte

B.S. Computer Science, Concentration in AI and Robotics

Charlotte, NC

Dec 2026

Coursework: Data Structures and Algorithms, Operating Systems and Networks, Introduction to AI, Linear Algebra, Discrete Structures of Software Engineering

Davidson-Davie Community College – Associate of Science (Dual Enrollment)

May 2023

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C, Swift, R, SQL

AI / ML: AI Agents, MCP, RAG, LLM Evaluation, PyTorch, Hugging Face Transformers, scikit-learn, Computer Vision, Depth Estimation

Backend / Data: FastAPI, REST APIs, WebSockets, PostgreSQL, SQLite, ChromaDB, Data Pipelines

Infrastructure: Linux, Bash, Git, Kubernetes, OpenShift, AWS, CUDA

Tools / Frontend: React, Ollama, Playwright, VS Code Extensions